

SKYLARK DRONES — EXECUTIVE BI & AGENT PLATFORM

Technical Decision Log & Architectural Rationale | Strict 2-Page Executive Summary

1. Tech Stack Selection & Competitor Trade-off Analysis

Layer	Chosen Tech	Competitor Evaluated	Architectural Rationale & Trade-off Decisions
Frontend UI	React 18 + Vite + Tailwind CSS + Recharts	Streamlit / Retool	Streamlit is rapid for simple scripts but lacks granular multi-tab navigation, custom conversational drawers, and instant 1-click visual switching. React provides responsive enterprise modularity and sub-second rendering.
Backend API	Python FastAPI + Pandas + Uvicorn	Flask / Django	FastAPI provides asynchronous execution, automatic Swagger OpenAPI documentation, native Pydantic validation, and high-performance in-memory Pandas aggregations without Django's monolithic overhead.
AI / LLM Brain	Groq Cloud API (Llama 3.3 70B)	Gemini / Ollama	Gemini free tiers suffer severe rate limits (HTTP 429) on high-frequency BI calculations. Ollama has 5-10s local latency. Groq's LPU acceleration delivers sub-300ms inference with large context windows.
Database & Persistence	Supabase (PostgreSQL + REST Client)	Self-hosted Postgres / LocalStorage	Eliminates infrastructure overhead while providing managed relational tables, instant REST APIs, and multi-session persistence with client-side fallback on maintenance.
Data Pipeline	Monday.com GraphQL v2 + Resilient Cache	Static Hardcoded CSV	Live GraphQL v2 schema discovery connects directly to active Deal Funnel and Work Order boards, with automatic fallback to a normalized local cache to ensure 100% evaluation uptime.

2. Data Normalization & 3-Tier Missing Value Resilience Framework

Tier Level	Mathematical / Empirical Strategy	Application in Skylark Datasets
Tier 1: Inter-Column Mathematical Calculation	Exact intra-row arithmetic and standard lead time formulas.	• GST Separation: $\text{Amount Excl GST} = \text{Amount Incl GST} / 1.18$ • Deterministic AR: $\max(0, \text{Billed} - \text{Collected Cash})$ • Sales Cycle: $\text{Tentative Close} = \text{Created Date} + 60 \text{ days}$
Tier 2: Pattern Matching & Entity Cross-Linking	Entity cross-referencing between sales opportunities and operational contracts.	• Deals missing monetary value search Work Orders by Deal Name to copy confirmed contract amounts. • Stage-Based Win Rates: Lead=10%, Qualified=25%, Proposal=60%, Negotiations=80%, Won=100%.
Tier 3: Statistical Median Fallback	Median imputation across sector clusters to prevent outlier distortion.	• Deals lacking entity matches inherit historical Sector Median Deal Size (Mining: ₹12.8L, Energy: ₹18.5L), with global portfolio median fallback.

3. Executive UX & Platform Capabilities

Feature Innovation	Mechanism Implemented	Strategic Business Benefit
Executive KPI Ribbon & Risk Radar	Persistent top ribbon tracking Open Pipeline, Weighted Value, Won Bookings, AR Exposure, and Software Attach rate.	Eliminates prompt fatigue; provides founders instant passive visibility into cash flow risks and overdue deliverables.
1-Click Multi-View Visualizer	Dynamic mode switcher (Bar Chart, Area Timeline Trend, Donut Pie Share, Data Grid, Metric Cards).	Allows executives to pivot between high-level visual distributions and granular audit records without sending follow-up prompts.
Interactive Board Explorer	Dedicated top tab with live multi-dimensional filters (Sector, Stage, Quarter, Search) and 1-click CSV Export.	Empowers operations and sales teams to explore, slice, and export raw CRM data without touching complex SQL queries.
Full CRUD Chat Management	Sidebar thread creation, inline title renaming, prompt editing, and message truncation (Undo Last).	Allows topic-isolated research (e.g. Q3 Cash Flow vs Energy Pipeline) with cross-device cloud persistence.
Dual Chatbot Architecture	Core Executive Monday.com BI Agent + Floating App Navigator Widget (bottom-right).	The App Navigator assists users in finding features and running health audits, while the core BI Agent handles deep analytical inquiries.